

LSTM-BASED APPROACH FOR GENERATOR STATE PREDICTION IN TRANSIENT STABILITY ASSESSMENT AND CONTROL USING TRANSIENT ENERGY FUNCTION

Presented By:

Bhupesh Jaishi (078BEL026)
Binod Kumar Sah (078BEL030)
Bishal Chaurasiya (078BEL031)
Chandan Sah (078BEL032)

DEPARTMENT OF ELECTRICAL ENGINEERING
IOE, PULCHOWK CAMPUS

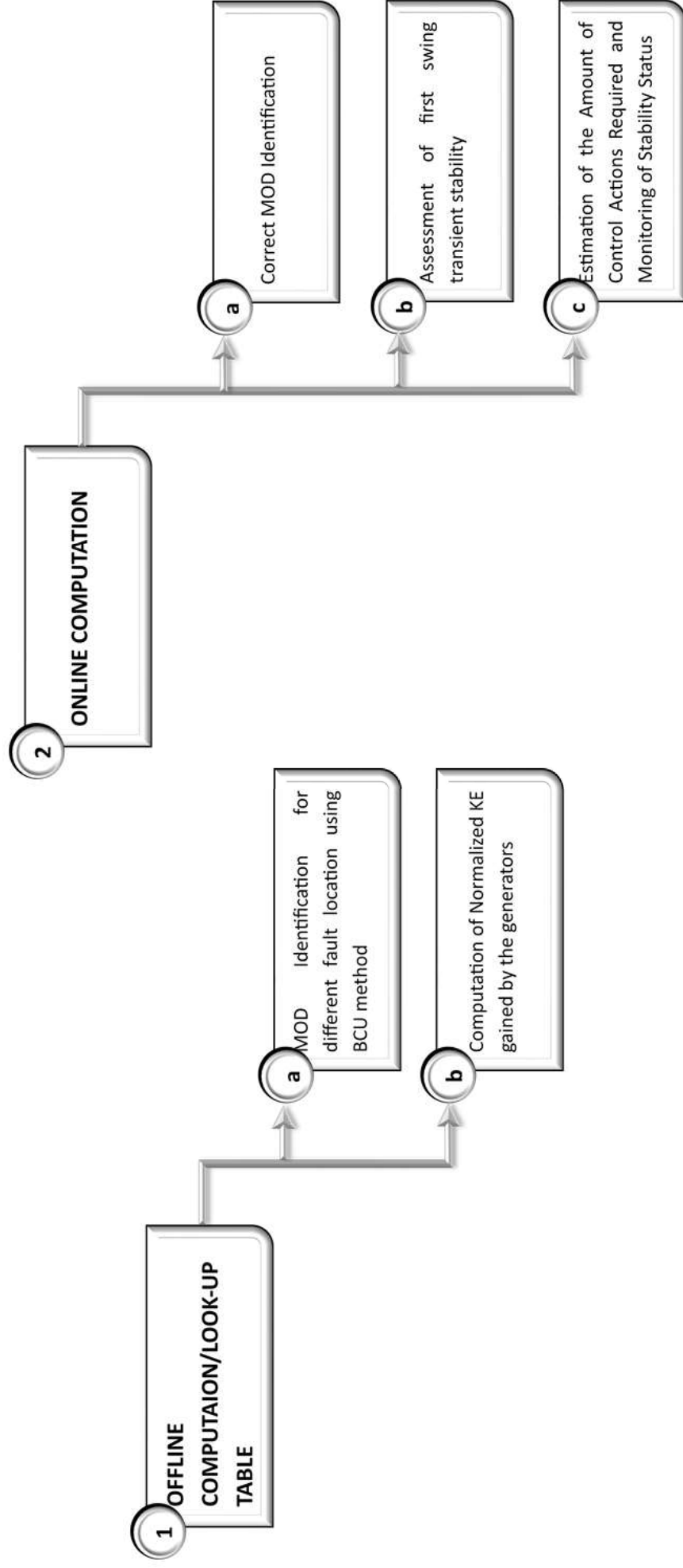
Introduction

- Modern power systems are becoming more dynamic and less predictable due to high renewable penetration and reduced inertia.
- After a severe disturbance, operators must determine transient stability and decide on emergency control actions within milliseconds.
- Traditional methods like full time-domain simulation or classical Transient Energy Function (TEF) analysis are too slow for real-time use.
- To address this, we develop a fast and intelligent hybrid framework that combines:
 - Offline TEF-based stability boundary computation, and
 - Online PMU-driven assessment enhanced with LSTM neural networks for rapid rotor-angle prediction.
- Our goal: enable ultra-fast, accurate, and reliable transient stability assessment and emergency control for modern power systems.

Objectives

- To develop Transient Energy Function (TEF) framework as discussed in our base paper
- To develop and implement an LSTM (Long Short-Term Memory) neural network model for accurate and real-time prediction of generator angles and speeds
- To demonstrate the applicability of the enhanced TEF method on the IEEE 39-bus test system

METHODOLOGY



1. Offline Computation/Look-up Table

a. MOD Identification for different fault location MOD is obtained using BCU Method.

1. Tcr is calculated using BCU method
2. Time domain simulation is done for a fault duration slightly higher than Tcr.
3. Detection of peak of the potential energy and generator ranking descending order of the angles.
4. Taking top machines as MOD and calculate CCT using MOD technique.
5. machine groups with minimum error between time domain simulating CCT and TEF CCT taken as the MOD.

b. Computation of normalized KE gained by the generators:

$$KE_X = [ke_{1x} \ ke_{2x} \ \dots \ ke_{ix} \ \dots \ ke_{mx}]$$

$$\overline{KE_X} = \frac{[\overline{ke_{1x}} \ \overline{ke_{2x}} \ \dots \ \overline{ke_{ix}} \ \dots \ \overline{ke_{mx}}]}{[ke_{1x} \ ke_{2x} \ \dots \ ke_{ix} \ \dots \ ke_{mx}]} = \frac{\sum_{i=1}^m \overline{ke_{ix}}}{\sum_{i=1}^m ke_{ix}}$$

2. Online Computation

a. Correct Mode Identification

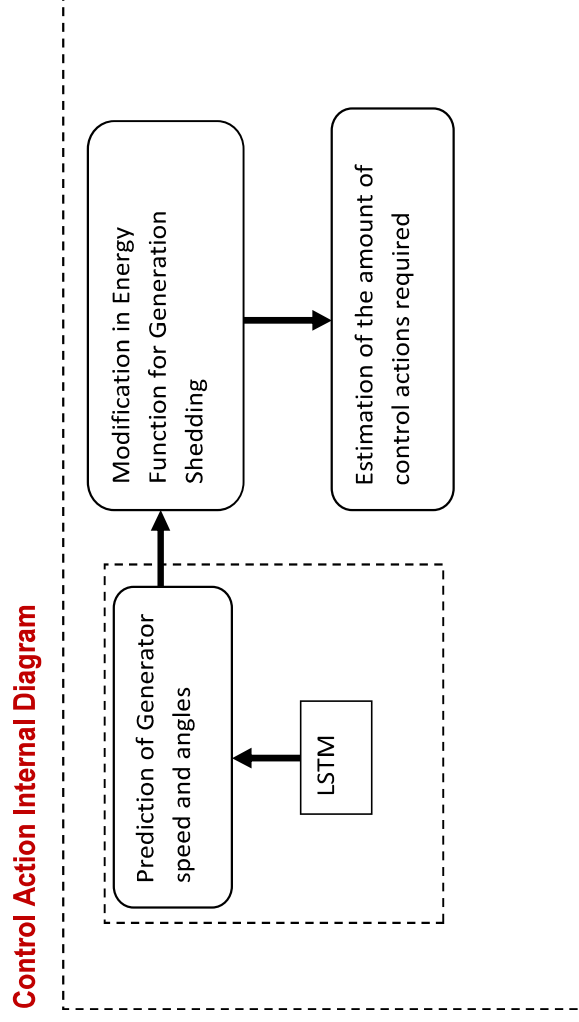
- Step 1 : Calculation of KE
- Step 2 : Ranking of Probable MODs
- Step 3 : Calculation of Post-Fault SEP
- Step 4 : Calculation of CUEP
- Step 5 : Identification of correct MOD

b. Assessment of first swing transient stability

$$\Delta V_{cl} = V_{cr} - V_{cl} \quad \begin{array}{l} >0, \text{ Stable} \\ <0, \text{ Unstable} \end{array}$$

c. Estimation of Amount of Control Actions Required and Monitoring of Stability Status

- Step 1 : Prediction of generator speed and angles[LSTM]
- Step 2 : Modification in Energy Function for Generation Shedding
- Step 3 : Estimation of the amount of Control action required



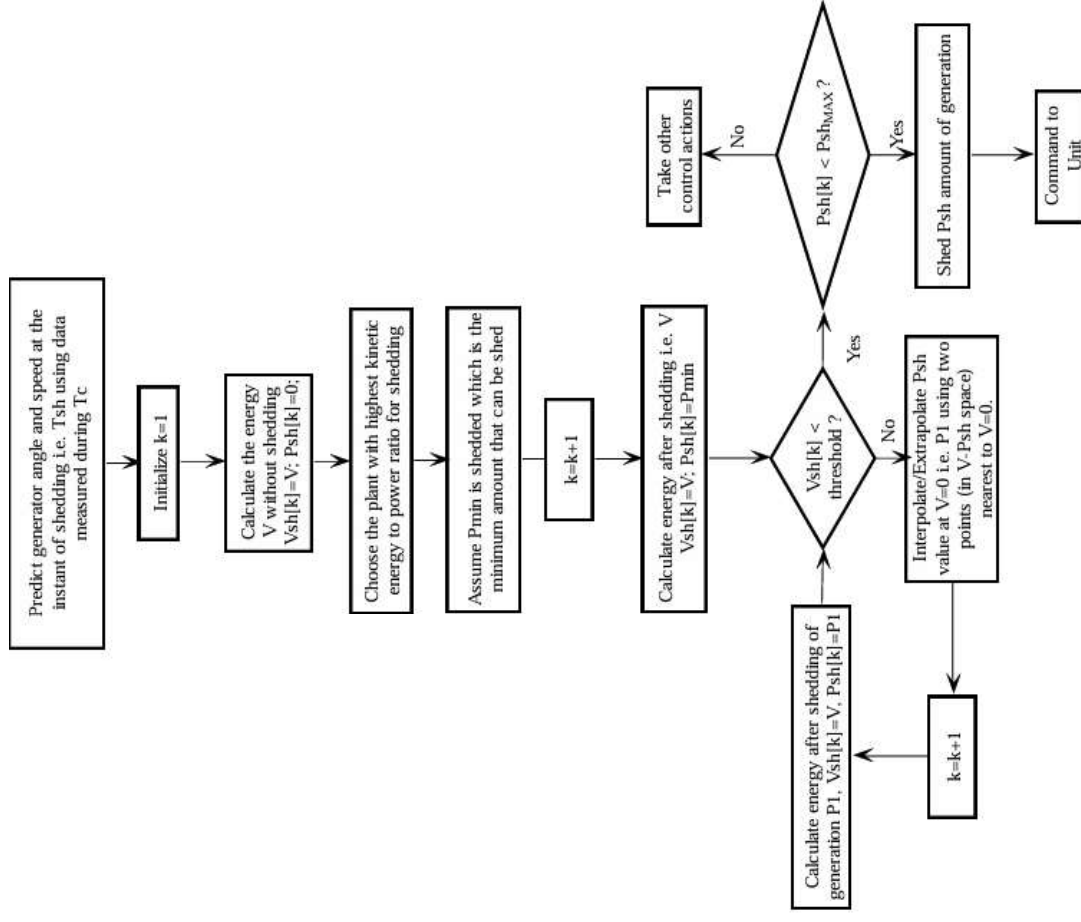


Fig. 1: Flowchart for generation shedding

Transient Energy Function(TEF)

It describes the total system transient energy for the post disturbance system.

Based on the data obtained from PMU the TEF is calculated using.

$$V(\omega_i, \theta_i) = V_{KE} + V_{PE} = \{0.5 M_{eq} \omega_{eq}^2\} - \left\{ \sum_{i=1}^m P_i (\theta_i - \theta_i^s) + \sum_{i=1}^{m-1} \sum_{j=i+1}^m [C_{ij} (\cos \theta_{ij} - \cos \theta_{ij}^s) + D_{ij} \frac{\theta_i - \theta_i^s + \theta_j - \theta_j^s}{\theta_{ij} - \theta_{ij}^s} (\sin \theta_{ij} - \sin \theta_{ij}^s)] \right\}$$

References

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Thank You!